



$$y^2 + xy - 4x - 9y + 20 = 0$$

$$y^2 + 2y \left(\frac{x-9}{2} \right) + \left(\frac{x-9}{2} \right)^2 - \frac{(x-9)^2}{2}$$

$$\left(y + \frac{(x-9)}{2} \right)^2 - \frac{(x-9)^2}{4} - 4x + 20 = 0$$

$$y = 20 - 4x$$

$$(y-4)(y+x-5)$$

$$\left(y + \frac{x-9}{2} \right)^2 - \frac{(x^2 - 18x + 81)}{4} - 4x + 20$$

$$4 \left(y + \frac{x-9}{2} \right)^2 - x^2 + 18x - 81 + 18x + 80 = 0$$

$$(2y + x - 9)^2 - x^2 + 2x - 1 = 0$$

$$(2y + x - 9)^2 - (x - 1)^2 = 0$$

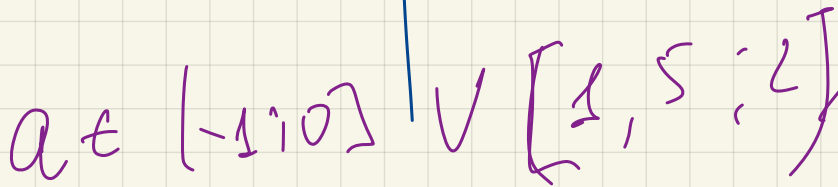
$$(2y + x - 9 - x + 1)(2y + x - 9 + x - 1) = 0$$

$$(2y - 8)(2y + 2x - 10) = 0$$

$$(y-4)(y+x-5)=0$$

$$y = ax + \underline{b}$$

1 pen



$$\begin{cases} \underline{6x^2 - 5xy + y^2 + x - y - z = 0} \\ y = ax - 5 \end{cases}$$

$$y = ax - 5$$

I play

$$y^2 - 2y \left(\frac{1+5x}{2} \right) + \left(\frac{1+5x}{2} \right)^2 - \left(\frac{1+5x}{2} \right)^2$$

$$\left(y - \frac{1+5x}{2} \right)^2 - \frac{(1+25x^2+10x)}{4} + x + 6x^2 - 7 = 0$$

$$\left(y - \frac{1+5x}{2} \right)^2 - 1 - 25x^2 - 10x + 24x^2 + 4x - 8 = 0$$

$$(2y - 5x - 1)^2 - x^2 - 6x - 9 = 0$$

$$(2y - 5x - 1)^2 - (x^2 + 6x + 9) = 0$$

$$(x+3)^2$$

$$(2y - 5x - 1)^2 - (x+3)^2 = 0$$

$$2^2 \left(y - \frac{1+5x}{2} \right)^2 = \left(2 \left(y - \frac{1+5x}{2} \right) \right)^2 =$$

$$\left\{ \begin{array}{l} x^2 + xy - 2x - 5y - 15 = 0 \\ y = ax \end{array} \right.$$

$$(x+y+3) \mid x-5$$

$$(2x-10-2y)(x+6)$$

$$x^2 + 2x \left(\frac{y-2}{2} \right) + \left(\frac{y-2}{2} \right)^2$$

$$\left(x + \frac{y-2}{2} \right)^2 = \frac{y^2 - 4y + 4}{4} - 5y - 15$$

$$\begin{aligned} (2x + y - 2)^2 &= y^2 + 4y + 4 - 2xy - 60 \\ &= y^2 - 2xy - 56 \\ &= (y-8)^2 \end{aligned}$$

$$(2x + \cancel{y} - 2 - \cancel{y} + 8)(2x + y - 2 + y + 8)$$

$$(2x - 10)(2x + 2y + 6) = 0$$

$$y(y-7) = xy - 5(x+2)$$

$$y^2 - 7y = xy - 5x - 10$$

$$y^2 - 7y - xy + 5x + 10 = 0$$

$$y^2 - y(7+x) + \left(\frac{7+x}{2}\right)^2 - \left(\frac{7+x}{4}\right)^2$$

$$\left(y - \frac{7+x}{2}\right)^2 - \left(\frac{7^2 + x^2 + 14x}{4}\right) + 5x + 10$$

$$(2y - 7 - x)^2 - 49 - x^2 - 14x + 20x + 40$$

$$-x^2 + 6x - 9$$

$$-(x^2 - 6x + 9)$$

$$(x-3)^2$$

$$\Delta/3$$